Time Series Exercises: Exponential Smoothing

Honours in Statistical Sciences

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Question 1

1. For the `marathon` data set in library `fpp2` write your own functions to implement simple exponential smoothing. You need to calculate error sums of squares and estimate the parameters α and ℓ_0. The function should take arguments y (the time series), alpha (the smoothing parameter α) and level (the initial level ℓ_0). It should return the forecast of the next observation in the series. Does it give the same forecast as `ses()`?
2. Use simple exponential smoothing to obtain a point forecast for 2017, and a prediction interval for 2017. With and without the `ses()` function. The estimates should be exactly the same.
3. Plot the time series and add fitted values.
4. Obtain an estimate for σ.

Question 2

Consider the oil series (annual oil production in Saudi Arabia, in library fpp2). Use the ses() function in R to find the optimal values of α and ℓ_0, and generate forecasts for the next four years. Compute a 95% prediction interval for the first forecast using $\hat{y} \pm 1.96 \times s$, where s is the standard deviation of the residuals. Compare your interval with the interval produced by R.

Question 3

Modify your function from the previous exercise to return the sum of squared errors rather than the forecast of the next observation. Then use the optim() function to find the optimal values of α and ℓ_0. Use this on a subset of the oil data: 1996-2013. Do you get the same values as the `ses()` function?

Question 4

a.) Data set `books` contains the daily sales of paperback and hardcover books at the same store. The task is to forecast the next four days' sales for paperback and hardcover books. Plot the series and discuss the main features of the data. Use the `ses()` function to forecast each series, and plot the forecasts. Compute the RMSE values for the training data in each case.

b.) Now apply Holt's linear method to the paperback and hardback series and compute four-day forecasts in each case. Compare the RMSE measures of Holt's method for the two series to those of simple exponential smoothing in a.). Remember that Holt's method is using one more parameter than SES. Discuss the merits of the two forecasting methods for these data sets.

c.) Compare the forecasts for the two series using both methods. Which do you think is best?

d.) Calculate a 95% prediction interval for the first forecast for each series, using the RMSE values and assuming normal errors. Compare your intervals with those produced using `ses()` and `holt()`.

Question 5

Simulate data using the state-space model formulation of simple exponential smoothing (and Holt's exponential smoothing if you want).

Then estimate the parameters using optim, and the R functions.

Explore

Problem 1: Forecasting Error Variance

Explore the different ways to forecast with a simple exponential smoothing model:

- bootstrap
- assuming normal errors (standard way)
- the options provided by the forecast function
- a Bayesian approach

Take particular note of the error variance estimates, exactly how they are calculated and how they compare. Distinguish between confidence intervals, prediction intervals, and which sources of uncertainty are captured by each of these intervals.

Summarize your results and point out similarities, differences, advantages, limitations. Reflect on these results.

You can use any data sets, or simulate the data so that you know the true values.

Problem 2: Bayesian Exponential Smoothing

Write functions to estimate parameters and predict (always with intervals) the three types of exponential smoothing approaches in a Bayesian framework.

You can use any data sets, or simulate the data so that you know the true values.

Reflect on these results.

Problem 3: Model Validation

This is a general problem in time series. It is relevant to all forecasting models and methods, not only exponential smoothing. You can use this to compare different methods.

The observed time series in the test set is only one realization from the stochastic process. You can calculate a metric such as RMSE on the test data for a particular model, or you can use other approaches.

Explore this problem and discuss in more detail. Suggest ways to compare models, which of these methods are preferable, and why. How is the time series setting different to a regression setting. What about machine learning algorithms? What is done in general?

Your exploration should include simulations to demonstrate the problem and a solution.

Your summary should explain the issue in some detail and discuss options and solutions.